

1. Identification

Product identifier	Calcium Indicator Liquid
Product code	R-0011L
Recommended use	Use as directed by manufacturer for purposes directly related to water testing.
Recommended restrictions	None known

Manufacturer/Importer/Supplier/Distributor information

Manufacturer

Company name	Taylor Technologies, Inc.		
Address	31 Loveton Circle Sparks, MD 21152 United States		
Telephone	(410) 472-4340	Monday–Friday, 8:00 a.m.–4:30 p.m.	
Website	www.taylortechnologies.com		
E-mail	Not available		
Emergency phone number	(800) 837-8548		

2. Hazard(s) identification

Physical hazards	Flammable liquids	Category 2
Health hazards	Eye damage/irritation	Category 2A
	Specific target organ toxicity, single exposure	Category 3 narcotic effects
	Specific target organ toxicity, single exposure	Category 3 respiratory tract irritation
Environmental hazards	Not currently regulated by OSHA. For additional information, refer to section 12 of the SDS.	

Label elements



Signal word	Danger
Hazard statement	Highly flammable liquid and vapor. Causes serious eye irritation. May cause respiratory irritation. May cause drowsiness or dizziness.

Precautionary statement

Prevention	Keep away from heat/sparks/open flames.-No smoking. Keep container tightly closed. Ground or bond container and receiving equipment. Use explosion-proof electrical/ventilating equipment. Use only non-sparking tools. Take precautionary measures against static discharge. Wear protective gloves/eye protection/face protection. Wash skin thoroughly after handling. Avoid breathing mist or vapor. Use only outdoors or in a well-ventilated area.
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Response	IF ON SKIN (OR HAIR): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing. IF EYE IRRITATION PERSISTS: Get medical advice/attention. Call a physician or poison control center if you feel unwell.
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	IN CASE OF FIRE: Use alcohol-resistant foam. Water fog. Carbon dioxide. Dry chemical powder, carbon dioxide, sand or earth may be used for small fires only.
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Storage	Store in a well-ventilated place. Keep container tightly closed. Keep cool. Store locked up.
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Disposal	Dispose of contents/container in accordance with local/regional/national/international regulations.
Hazard(s) not otherwise classified	None
Supplemental information	None

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
Triethanolamine	2,2',2''-Nitrilotriethanol; Tris(2-hydroxyethyl)amine	102-71-6	75–80
Isopropanol	Dimethyl carbinol; 2-Propanol; Isopropyl alcohol	67-63-0	20–25
Calcon	1-(2-Hydroxy-1-naphthylazo)-2-naphthol-4-sulfonic acid sodium salt; Mordant black 17	2538-85-4	0.1–5

4. First-aid measures

Inhalation	Move to fresh air. Give oxygen or artificial respiration if needed. Get medical attention if you feel unwell.
Skin contact	Immediately flush skin with running water for at least 20 minutes. Immediately take off all contaminated clothing. Get medical attention if you feel unwell. Wash contaminated clothing before reuse.
Eye contact	Immediately flush eyes with plenty of water for at least 20 minutes. Remove contact lenses if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	Call a physician or poison control center immediately. Rinse mouth. Never give anything by mouth to a person who is unconscious or is having convulsions. Do NOT induce vomiting unless directed by physician. If vomiting occurs, keep head low so that stomach content does not get into the lungs.
Most important symptoms/effects, acute and delayed	Direct skin contact may cause slight or mild transient irritation. Symptoms may include redness and itching. Direct eye contact may cause slight or mild transient irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Inhalation of mists can cause respiratory irritation. Symptoms may include coughing and breathing difficulties. May cause drowsiness or dizziness. Symptoms may include pain, headache, nausea, vomiting, dizziness, drowsiness, and other central nervous system effects. Ingestion may cause gastrointestinal irritation, nausea, vomiting, diarrhea, as well as depression of the central nervous system.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically.
General information	This product is a CNS depressant. Ensure medical personnel are aware of the material(s) involved and take precautions to protect themselves.

5. Firefighting measures

Suitable extinguishing media	Alcohol-resistant foam. Water fog. Carbon dioxide. Dry chemical powder, carbon dioxide, sand or earth may be used for small fires only.
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Vapors may form explosive mixtures with air. Vapors may travel considerable distance to a source of ignition and flash back. This product is a poor conductor of electricity and can be electrostatically charged. If sufficient charge is accumulated, ignition of flammable mixtures can occur. To reduce potential static discharge, use proper bonding and grounding procedures. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Self-contained breathing apparatus and full protective clothing must be worn in case of fire.

Firefighting equipment/instructions

Firefighters should wear full protective gear. Evacuate the area promptly. Fight fire from upwind to avoid exposure to combustion products. Cool containers/tanks with water spray. Do not get water inside container. Move containers from fire area if it can be done without risk. Prevent fire-extinguishing water from contaminating surface water or the ground water system.

Specific methods

Use standard firefighting procedures and consider the hazards of other involved materials.

General fire hazards

Flammable liquid and vapor. This material may be ignited by heat, sparks, flames, or other sources of ignition (e.g., static electricity, pilot lights, or mechanical/electrical equipment). Vapors are heavier than air and may spread along floors. Vapors may travel considerable distance to a source of ignition and flash back.

Hazardous combustion products

Carbon oxides. Nitrogen oxides. Peroxides. Other irritating fumes and smoke.

6. Accidental release measures

Personal precautions, protective equipment, and emergency procedures

Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Keep out of low areas. Wear appropriate protective equipment and clothing during cleanup. Do not breathe mist or vapor. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protective equipment, refer to section 8 of the SDS.

Methods and materials for containment and cleaning up

Ventilate the contaminated area. Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Take precautionary measures against static discharge. Use only non-sparking tools.

Large Spills: Dike the spilled material where this is possible. Stop leak if it can be done without risk. Absorb spillage to prevent material damage. Absorb in vermiculite, dry sand or earth, and place into containers. Prevent entry into waterways, sewer, basements, or confined areas. Following product recovery, flush area with water.

Small Spills: Absorb spillage with noncombustible, absorbent material. Clean surface thoroughly to remove residual contamination.

Never return spills to original containers for reuse. For waste disposal, refer to section 13 of the SDS. Contaminated absorbent material may pose the same hazards as the spilled product.

In the event of a spill or accidental release, notify relevant authorities in accordance with all applicable regulations.

Environmental precautions

Avoid discharge into drains, watercourses, or onto the ground.

7. Handling and storage

Precautions for safe handling

Vapors may form explosive mixtures with air. Keep away from sources of ignition. NO SMOKING. Do not handle, store, or open near an open flame, sources of heat or sources of ignition. All equipment used when handling the product must be grounded. Use non-sparking tools and explosion-proof equipment. Do not breathe mist or vapor. Do not get in eyes, on skin, or on clothing. Do not taste or swallow. Provide adequate ventilation. Wear appropriate personal protective equipment. For personal protective equipment, refer to section 8 of the SDS. Keep away from incompatibles. Observe good industrial hygiene practices. Label containers appropriately.

Conditions for safe storage, including any incompatibilities

Store locked up. Store in vented containers. Keep away from heat, sparks, and open flames. This material can accumulate static charge which may cause a spark and become an ignition source. Prevent electrostatic charge buildup by using common bonding and grounding techniques. Store in a well-ventilated place. Store in a cool, dry place out of direct sunlight. Store away from incompatible materials (refer to section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

U.S. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value	Form
Isopropanol (CAS 67-63-0)	PEL	980 mg/m ³	Not applicable
		400 ppm	Not applicable

U.S. ACGIH Threshold Limit Values

Components	Type	Value	Form
Isopropanol (CAS 67-63-0)	STEL	400 ppm	Not applicable
	TWA	200 ppm	Not applicable
Triethanolamine (CAS 102-71-6)	TWA	5 mg/m ³	Not applicable

U.S. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value	Form
Isopropanol (CAS 67-63-0)	STEL	1225 mg/m ³	Not applicable
		500 ppm	Not applicable
	TWA	980 mg/m ³	Not applicable
		400 ppm	Not applicable

Biological limit values**U.S. ACGIH Biological Exposure Indices**

Components	Value	Determinant	Specimen	Sampling Time
Isopropanol (CAS 67-63-0)	40 mg/L	Acetone	Urine	Not available

Appropriate engineering controls

Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level.

Individual protection measures, such as personal protective equipment**Eye/face protection**

Wear safety glasses with side shields (or goggles) and a face shield. Provide an emergency eyewash fountain and quick-drench shower in the immediate work area.

Skin protection**Hand protection**

Wear appropriate chemical-resistant gloves. Advice should be sought from glove suppliers.

Other

Wear appropriate chemical-resistant clothing.

Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment. Use a NIOSH/MSHA approved respirator if there is a risk of exposure to fumes at levels exceeding the exposure limits. Advice should be sought from respiratory protection suppliers.

Thermal hazards

When necessary, wear appropriate thermal protective clothing.

General hygiene considerations

Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking and/or smoking. Routinely wash work clothing and protective equipment to remove contamination. Avoid breathing mist or vapor.

9. Physical and chemical properties**Appearance****Physical state**

Liquid

Form

Liquid

Color

Dark purple to dark blue

Odor

Alcohol-like

Odor threshold

Not available

pH

10.3

Melting point/freezing point

Not available

Initial boiling point and boiling range

500-600°F (260-315.56°C)

Flash point

64.0°F (17.8°C) Closed Cup

Evaporation rate

Not available

Flammability (solid, gas)

Flammable

Upper/lower flammability or explosive limits**Flammability limit, lower (%)**

2%

Flammability limit, upper (%)

12%

Explosive limit, lower (%)

Not available

Explosive limit, upper (%)

Not available

Vapor pressure

Not available

Vapor density

2

Triethanolamine (CAS 102-71-6)

Acute

Dermal

LD ₅₀	Rabbit	Not available
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Inhalation

LC ₅₀	Rat	Not available
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Oral

LD ₅₀	Mouse	5846 mg/kg
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	Rabbit	2200 mg/kg
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Deionized water (CAS 7732-18-5)

Acute

Dermal

LD ₅₀	Rabbit	Not available
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Inhalation

LC ₅₀	Rat	Not available
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Oral

LD ₅₀	Rat	>89840 mg/kg
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Skin corrosion/irritation	Causes skin irritation
Serious eye damage/eye irritation	Causes severe eye irritation
Respiratory sensitization	Not expected to be a respiratory sensitizer
Skin sensitization	Not expected to be a skin sensitizer
Germ cell mutagenicity	Not expected to be mutagenic
Carcinogenicity	This product is not considered to be a carcinogen by IARC, NTP, OSHA, or U.S. ACGIH.

IARC Monographs. Overall Evaluation of Carcinogenicity

Triethanolamine (CAS 102-71-6)	3 Not classifiable as to carcinogenicity to humans
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OSHA Specifically Regulated Substances (29 CFR 1910.1001-1096)

Not regulated

Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
Specific target organ toxicity, single exposure	May cause drowsiness or dizziness. May cause respiratory irritation.
Specific target organ toxicity, repeated exposure	Not classified as a specific target organ toxicity – repeated exposure
Aspiration toxicity	Not expected to be an aspiration hazard
Chronic effects	Frequent or prolonged contact may dry the skin, leading to discomfort and dermatitis. Frequent or prolonged inhalation of fumes or vapors may cause chronic lung conditions such as bronchitis. Frequent or prolonged overexposure may affect the kidneys.

12. Ecological information

Ecotoxicity	This product is not classified as environmentally hazardous; however, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.
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Components	Species	Test Results
Isopropanol (CAS 67-63-0) – Aquatic		
Acute		
Crustacea		
EC ₅₀	Water flea (<i>Daphnia magna</i>)	1400 mg/L, 48 hours
LC ₅₀	Fathead minnow (<i>Pimephales promelas</i>)	9640 mg/L, 96 hours
Chronic		
Crustacea		
NOEC	Water flea (<i>Daphnia magna</i>)	30 mg/L, 21 days

Triethanolamine (CAS 102-71-6) – Aquatic

Acute

Algae

EC₅₀ Green algae (*Desmodesmus subspicatus*) 512 mg/L, 72 hours

Crustacea

EC₅₀ Water flea (*Ceriodaphnia affinis*) 609.88 mg/L, 48 hours

Chronic

Crustacea

NOEC Water flea (*Daphnia magna*) 16 mg/L, 21 days

Persistence and degradability Not available

Bioaccumulative potential

Partition coefficient n-octanol / water (log K_{ow})

Isopropanol 0.05

Triethanolamine -1

Bioconcentration factor (BCF)

Isopropanol 1

Mobility in soil High water solubility indicates a high mobility in soil.

Other adverse effects No other adverse environmental effects (e.g., ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Disposal instructions Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Dispose of contents/container in accordance with local/regional/national/international regulations.

Local disposal regulations Dispose in accordance with all applicable regulations.

Hazardous waste code The waste code should be assigned in discussion with the user, the producer, and the waste disposal company.

Waste from residues/unused products Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner (refer to Disposal instructions).

Contaminated packaging Empty containers should be taken to an approved waste-handling site for recycling or disposal. Since emptied containers may retain product residue, follow label warnings even after container is emptied.

14. Transportation information

DOT

UN number UN1993

UN proper shipping name Flammable liquids, N.O.S. (Isopropanol RQ = 100 lbs.)

Transport hazard class(es)

Class 3

Subsidiary risk Not listed

Label(s) 3

Packing group II

Special precautions for user Read safety instructions, SDS, and emergency procedures before handling.

Special provisions IB2, T7, TP8, TP28

Packaging exceptions 150

Packaging, non-bulk 202

Packaging, bulk 242

IATA

UN number UN1993

UN proper shipping name Flammable liquids, N.O.S. (Isopropanol)

Transport hazard class(es)

Class 3

Subsidiary risk Not listed

Packing group II

Environmental hazards Not listed

ERG code 3H

Special precautions for user Read safety instructions, SDS, and emergency procedures before handling.

Other information

Passenger and cargo aircraft Allowed

Cargo aircraft only	Allowed
IMDG	
UN number	UN1993
UN proper shipping name	Flammable liquids, N.O.S. (Isopropanol)
Transport hazard class(es)	
Class	3
Subsidiary risk	Not listed
Packing group	II
Environmental hazards	
Marine pollutant	No
EmS	F-E, S-E
Special precautions for user	Read safety instructions, SDS, and emergency procedures before handling.
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	This substance/mixture is not intended to be transported in bulk.
DOT	



IATA; IMDG



15. Regulatory information

U.S. federal regulations All components are on the U.S. EPA TSCA Inventory list.

TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)

Not regulated

CERCLA Hazardous Substance (40 CFR 302.4)

Isopropanol (CAS 67-63-0)

SARA 304 Emergency Release Notification

Not regulated

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1096)

Not regulated

Superfund Amendments and Reauthorization Act of 1986 (SARA)

Hazard categories

Immediate hazard – yes
Delayed hazard – no
Fire hazard – yes
Pressure hazard – no
Reactivity hazard – no

SARA 302 Extremely Hazardous Substance

Not regulated

SARA 311/312 Hazardous Chemical

Regulated

SARA 313 (TRI reporting)

Chemical name	CAS number	% by weight
Isopropanol	67-63-0	23

Other federal regulations

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAP)

Not regulated

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

Not regulated

Safe Drinking Water Act (SDWA)

Not regulated

U.S. state regulations

California Controlled Substances. CA Department of Justice (California Health and Safety Code Section 11100)

Not regulated

Massachusetts Right-to-Know Act

Isopropanol (CAS 67-63-0)

Triethanolamine (CAS 102-71-6)

New Jersey Worker and Community Right-to-Know Act

Isopropanol (CAS 67-63-0)

Triethanolamine (CAS 102-71-6)

Pennsylvania Worker and Community Right-to-Know Act

Isopropanol (CAS 67-63-0)

Triethanolamine (CAS 102-71-6)

Rhode Island Right-to-Know Act

Isopropanol (CAS 67-63-0)

California Proposition 65

California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65): This material contains a chemical known to cause cancer.

International inventories

Country(ies) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	yes
Canada	Domestic Substances List (DSL)	yes
Canada	Non-Domestic Substances List (NDSL)	no
China	Inventory of Existing Chemical Substances Produced or Imported in China (IECSC)	yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	yes
Europe	European List of Notified Chemical Substances (ELINCS)	no
Japan	Existing and New Chemical Substances (ENCS)	yes
Korea	Existing Chemicals List (ECL)	yes
New Zealand	New Zealand Inventory of Chemical (NZIoC)	yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA)	yes

*A "yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(ies).

A "no" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(ies).

16. Other information, including date of preparation or last revision

List of abbreviations

ACGIH: American Conference of Governmental Industrial Hygienists
AICS: Australian Inventory of Chemical Substances
CAA: Clean Air Act
CAS: Chemical Abstract Services
CERCLA: Comprehensive Environmental Response, Compensation, and Liability Act
CFR: Code of Federal Regulations
CSA: Canadian Standards Association
DEA: Drug Enforcement Agency
DOT: Department of Transportation
DSL: Domestic Substances List
EC: effective concentration
ECL: Existing Chemicals List
EINECS: European Inventory of Existing Commercial Chemical Substances
ELINCS: European List of Notified Chemical Substances

ENCS: Existing and New Chemical Substances
 EPA: Environmental Protection Agency
 HAP: hazardous air pollutants
 HMIS: Hazardous Materials Identification System
 HNOC: hazards not otherwise classified
 HPA: Hazardous Products Act
 HSDB: Hazardous Substances Data Bank
 IARC: International Agency for Research on Cancer
 IATA: International Air Transport Association
 IBC Code: International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk
 ICAO: International Civil Aviation Organization
 IECSC: Inventory of Existing Chemical Substances Produced or Imported in China
 IMDG: International Maritime Dangerous Goods
 IUCLID: International Uniform Chemical Information Database
 LC: lethal concentration
 LD: lethal dose
 MARPOL: marine pollution
 MSHA: Mine Safety and Health Administration
 NDSL: Non-Domestic Substances List
 NFPA: National Fire Protection Association
 NIOSH: National Institute of Occupational Safety and Health
 NOEC: no observable effect concentration
 NTP: National Toxicology Program
 NZIoC: New Zealand Inventory of Chemicals
 OECD: Organisation for Economic Co-operation and Development
 OEL: occupational exposure limits
 OSHA: Occupational Safety and Health Administration
 PEL: permissible exposure limits
 PICCS: Philippine Inventory of Chemicals and Chemical Substances
 PPE: personal protective equipment
 RCRA: Resource Conservation and Recovery Act
 RQ: reportable quantity
 RTECS: Registry of Toxic Effects of Chemical Substances
 RTK: right to know
 SARA: Superfund Amendments and Reauthorization Act
 SDS: Safety Data Sheet
 SDWA: Safe Drinking Water Act
 STEL: short-term exposure limit
 TLV: threshold limit values
 TSCA: Toxic Substances Control Act
 TWA: time-weighted average
 VOC: volatile organic compounds
 WEL: workplace exposure limit

Disclaimer

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